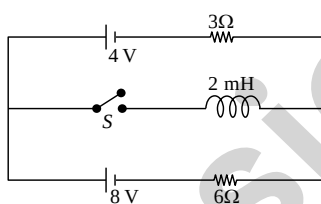
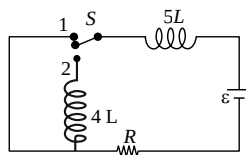


Section - A : MCQs with Single Option Correct

- Arrange the following electromagnetic radiations in the order of increasing energy :
A: Blue B: Yellow light
C: X-ray D: Radio wave
(A) C, A, B, D (B) B, A, D, C (C) D, B, A, C (D) A, B, D, C
- Two short bar magnets of length 1 cm each have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are placed on a horizontal table parallel to each other with their *N* poles pointing towards the South. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the mid-point *O* of the line joining their centres is close to : (Horizontal component of earth's magnetic induction is $3.6 \times 10^{-5} \text{ Wb/m}^2$)
(A) $3.6 \times 10^{-5} \text{ Wb/m}^2$ (B) $2.56 \times 10^{-4} \text{ Wb/m}^2$ (C) $3.50 \times 10^{-4} \text{ Wb/m}^2$ (D) $5.80 \times 10^{-4} \text{ Wb/m}^2$
- In the given circuit, switch is closed at time(*t*) = 0 find current through the inductor as a function of time *t*.



- (A) $\frac{4}{3}\left(1 - e^{-\frac{4000t}{3}}\right)$ (B) $\frac{8}{3}\left(1 - e^{-\frac{4000t}{3}}\right)$ (C) $\frac{4}{3}\left(1 - e^{-\frac{2000t}{3}}\right)$ (D) $\frac{8}{3}\left(1 - e^{-1000t}\right)$
4. In an LC oscillator, $L = 12 \text{ mH}$ & $C = 1.6 \mu\text{F}$ and $R = 1.5 \Omega$. Find the time in which max charge on capacitor falls to half of its initial value (approximately)
- (A) 5 ms (B) 11 ms (C) 17 ms (D) 23 ms
5. In the circuit shown in figure, the switch S is shifted to position 2 from position 1 at $t = 0$, having been in position 1 for a long time. The current in the circuit just after shifting of switch will be it (battery and both the inductors are ideal)



- (A) $\frac{4}{5} \frac{\varepsilon}{R}$ (B) $\frac{5}{4} \frac{\varepsilon}{R}$ (C) $\frac{5}{9} \frac{\varepsilon}{R}$ (D) $\frac{\varepsilon}{R}$
6. A long solenoid with 10.0 turns/cm and a radius of 7.00 cm carries a current of 20.0 mA. A current of 6.00 A exists in a straight conductor located along the central axis of the solenoid. At what radial distance (approx) from the axis-will the direction of the resulting magnetic field be at 45.0° to the axial direction ?
- (A) 4.22 cm (B) 4.77 cm (C) 4 cm (D) 2.77 cm

7. Two concentric coils each of radius equal to 2π cm are placed at right angles to each other. 3 A and 4 A are the currents flowing in each coil respectively. The magnetic induction in Wb/m^2 at the centre of the coils will be :
 $[\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am}]$
 (A) 12×10^{-5} (B) 10^{-5} (C) 5×10^{-5} (D) 7×10^{-5}

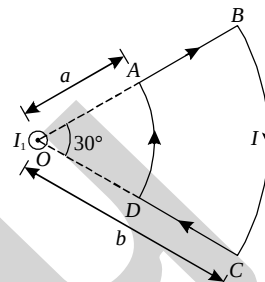
8. A current I flows along the length of an infinitely long, straight, thin walled pipe. Then :
 (A) The magnetic field at all points inside the pipe is the same, but not zero
 (B) The magnetic field is zero only on the axis of the pipe
 (C) The magnetic field is different at different points inside the pipe
 (D) The magnetic field at any point inside the pipe is zero

9. Due to the presence of the current I_1 at the origin :

(A) The magnitude of the net force on the loop is given by $\frac{I_1 I}{4\pi} \mu_0 \left[2(b-a) + \frac{\pi}{3}(a+b) \right]$

(B) The magnitude of the net force on the loop is given by $\frac{\mu_0 I I_1}{24ab} (b-a)$

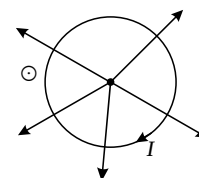
- (C) The forces on AB and DC are zero
 (D) The forces on AD and BC are zero



10. An electric field alone produces acceleration of 0.25×10^{14} on a charge particle. When this particle enters normally into a 1 T magnetic field at speed 100 m/s, this electric field makes this particles move on a straight line. The q/m ratio for this particle is :
 (A) 10^{12} C/kg (B) $0.25 \times 10^{12} \text{ C/kg}$ (C) $32 \times 10^{12} \text{ C/kg}$ (D) $4.8 \times 10^{12} \text{ C/kg}$

11. An insulating thin rod of length 1 m has linear charge density $\lambda = 0.5x$ on it where x is distance from one end of rod. The rod is rotated about an axis passing through the origin ($x = 0$) and perpendicular to the rod. If the rod makes 2 rotations per second, then the time averaged magnetic moment of the rod is :
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{8}$ (D) π

12. A circular loop of radius R , carrying current I is placed in a magnetic field. Calculate the magnetic force experience by the wire if the magnetic field (B) is uniform & acting perpendicular to plane of loop ? What would be the value of magnetic force if magnetic field is along radially outward direction & its strength is varying as $B = Ar$ where r is the distance measured from center of field which is coinciding with center of loop ?
 (A) $2\pi R \times A_0 \times I$, zero
 (B) zero, $2\pi R \times I \times Ar$
 (C) zero, $I \times A \times 2\pi R^2$
 (D) $2\pi R^2 AI$, zero



13. A closed loop made up of insulating material, having cross sectional area 10 cm^2 is placed in a varying magnetic field, which is increasing at the rate of 10 T/s . The plane of the loop is perpendicular to magnetic field. Calculate the induced emf & induced current ?
 (A) 0, 0 (B) 10^{-4} V , 10^{-6} V
 (C) 10^{-4} V , 0 (D) Information is insufficient

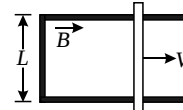
14. A metal rod is sliding on an U-shaped conductor as shown in figure. The resistance of the circuit is R , at any particular time during the motion of metal rod. Find out the rate at which energy is dissipated in the circuit at this moment ?

(A) $\frac{B^2 L^2 v^2}{3R}$

(B) $\frac{BLv}{3}$

(C) $\frac{B^2 L^2 v^2}{R}$

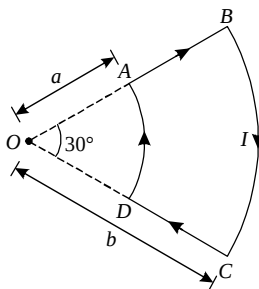
(D) $\frac{B^2 L^2 v^2}{2R}$



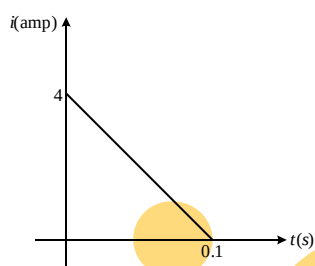
15. A conducting ring of radius R is rotated about its own axis, in a region where $\vec{B}$ is along the axis of ring. Calculate the emf induced between the center & a point on the periphery of the rim ?
 (A) $\frac{\omega BR^2}{2}$ (B) ωBR^2 (C) Zero (D) None
16. A bicycle tyre consisting of 124 metal spokes which are fastened between center & rim of tyre, is rotated about its own axis with angular velocity ω . The magnetic field $\vec{B}$ is present in the region which is acting along the axis of ring. Calculate the emf induced between the periphery & center of ring ?
 (A) $\frac{124B\omega R^2}{2}$ (B) $\frac{62\omega BR^2}{2}$ (C) $\frac{\omega BR^2}{2}$ (D) $\frac{73\omega BR^2}{2}$
17. When resonance is produced in a series LCR circuit, then which of the following is not correct ?
 (A) Current in the circuit is in phase with the applied voltage
 (B) Inductive and capacitive reactances are equal
 (C) If R is reduced, the voltage across capacitor will increase
 (D) Impedance of the circuit is maximum
18. When the rms voltages V_L , V_C and V_R are measured respectively across the inductor L , the capacitor C and the resistor R in a series LCR circuit connected to an AC source, it is found that the ratio $V_L : V_C : V_R = 1 : 2 : 3$. If the rms voltage of the AC sources is 100 V, the V_R is close to :
 (A) 50V (B) 70V (C) 90V (D) 100V
19. A sinusoidal voltage $V(t) = 100 \sin(500t)$ is applied across a pure inductance of $L = 0.02$ H. The current through the coil is:
 (A) $10 \cos(500t)$ (B) $-10 \cos(500t)$ (C) $10 \sin(500t)$ (D) $-10 \sin(500t)$
20. The AC voltage across a resistance can be measured using a :
 (A) Hot wire voltmeter (B) Moving coil galvanometer
 (C) Potential coil galvanometer (D) Moving magnet galvanometer

Section- B: INTEGER Answer Type Questions

21. In a experiment electrons are accelerated by applying a voltage of 2 kV. Calculate the radius (in mm) of the circular path of electron if subjected to a magnetic field of 50 mT.
22. A particle of mass 4×10^{-24} kg and charge same as that of electron enters a magnetic field of 1T in which it moves in a circular path of radius 2 cm. What electric field (in V/m) can be applied to make this particle move in a straight line path.
23. A horizontal overhead powerline is at a height of 4 m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground is $n \times 10^{-6}$ T southward. Calculate the value of n .
24. The magnitude of the magnetic field due to the loop $ABCD$ at the point O is given as $\frac{\mu_0 I(b-a)}{nab}$. Calculate the value of n .

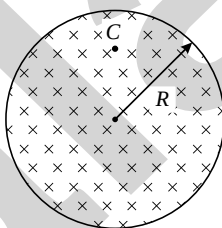


25. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to $2R$. A capacitance of capacitive reactance equal to R is added in series with L and R . The ratio of the new power factor to the old one is $\sqrt{\frac{n}{2}}$. Find n .
26. A bar magnet of length 6 cm has a magnetic moment of 4 JT^{-1} . The strength of magnetic field at a distance of 200 cm from the centre of the magnet along its equatorial line is $n \times 10^{-8}$ tesla. Find n .
27. When magnetic flux through a coil is changed, the variation of induced current in the coil with time is as shown in graph. If resistance of coil is 10Ω , then the total change in flux (in Weber) of coil will be :



28. Figure shows a uniform magnetic field of induction B confined to a cylindrical volume of radius R . B is increasing at a constant rate of 100 T/s. What is the instantaneous acceleration (in m/s^2) experienced by a charged particle placed at

C distant r from centre. Assume $r = 5 \text{ m}$. (Given $\left[\frac{q}{m} = 16 \times 10^{-3} \frac{\text{C}}{\text{kg}} \right]$)



29. Two circular coils P and Q are made from similar wires. The radius of Q is twice the radius of P . Find the ratio of potential difference across coil Q to the potential difference across coil P so that magnetic induction at their centres is same.
30. The magnetic field inside a conducting wire varies with distance r from its axis as $B = Kr^4$. If current density in the wire is $j = \frac{aKr^3}{\mu_0}$, find the value of a .

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JEE Main Level Practice Test-15

for JEE & NEET Aspirants

Topic : MAGNETIC EFFECTS
ANSWER KEY
Section - A : MCQs with Single Option Correct

- | | | | |
|---------|---------|---------|---------|
| 1. (C) | 2. (B) | 3. (D) | 4. (B) |
| 5. (C) | 6. (B) | 7. (C) | 8. (D) |
| 9. (D) | 10. (B) | 11. (B) | 12. (B) |
| 13. (C) | 14. (C) | 15. (C) | 16. (C) |
| 17. (D) | 18. (C) | 19. (B) | 20. (A) |

Section- B: INTEGER Answer Type Questions

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|---------|-----------|---------|----------|
| 21. [3] | 22. [800] | 23. [5] | 24. [24] |
| 25. [5] | 26. [5] | 27. [2] | 28. [4] |
| 29. [4] | 30. [5] | | |

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